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Excercise 1-1

1. a. Integers: any whole number.
   b. digits: a number that makes up others (0-9)
   c. even numbers: any number divisible without remainder by 2.
   d. positive numbers: any number above 0.
   e. negative numbers: any number below 0.
   f. rational numbers: a number that can be expressed by a ratio of exactly 2 integers.
   g. irrational numbers: a number that cannot be expressed by a ratio of exactly 2 integers.
   h. imaginary numbers: the square root of any negative number.
   i. real numbers: numbers that have points on a numberline.
   j. natural numbers: positive integers (also known as counting numbers).
   k. counting numbers: any positive integer.
   l. transcendental numbers: 2 a number that cannot be represented as the root of an integer.
2. a. 100
   b. 4
   c. 6
   d. 10
   e. -10
   f. $\frac{4}{5}$
   g. $\sqrt{-10}$
   h. $\sqrt{-40}$
   i. 3
   j. 10
   k. 67
   l. π
3. See other page
4. counting numbers
5. natural (or counting) numbers
6. real numbers contain all but imaginary
7. yes because they can be represented with two integers ($\frac{1}{2}$ from example)
8. yes because they can be represented with two integers ($\frac{1}{3}$ for example)
9. zero or 0